

Disentangling effects of dispersal, environment and anthropogenic barriers on functional connectivity in aquatic systems

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Abstract

Disentangling the roles of structural landscape factors and animal movement behaviour can present challenges for practitioners managing landscapes to maintain functional connectivity and achieve conservation goals. We used a landscape genetics approach to combine robust demographic, behavioural and genetic datasets with spatially explicit simulations to evaluate the effects of anthropogenic barriers (dams, culverts) and natural landscape resistance (gradient, elevation) affecting dispersal behaviour, genetic connectivity and genetic structure in a resident population of Westslope Cutthroat Trout (*Oncorhynchus clarkii lewisi*). Analyses based on 10 years of sampling effort revealed a pattern of restricted dispersal, and population genetics identified discrete population clusters between distal tributaries and the mainstem stream and no structure within the mainstem stream. Demogenetic simulations demonstrated that, for this population, the effects of existing anthropogenic barriers on population structure are redundant with effects of restricted dispersal associated with the underlying environmental resistance. Our approach provides an example of how extensive field sampling combined with landscape genetics can be incorporated into spatially explicit simulation modelling to explore how, together, movement ecology and landscape resistance can be used to inform decisions around restoration and connectivity.

KEYWORDS

CDMetaPOP, computer simulations, dispersal behaviour, habitat fragmentation, landscape connectivity, riverscape genetics

1 | INTRODUCTION

Functional connectivity of populations is influenced by landscape structure that facilitates or impedes animal movement and by the behavioural capacity of individuals to move across that landscape (Bélisle, 2005). Understanding how landscape structure and animal behaviour interact to influence functional connectivity is essential for understanding the status of populations and planning for their conservation. A decline in functional connectivity can result in population isolation, loss of genetic diversity due to genetic drift and lack of genetic connectivity, and an increased likelihood of extinction (Allendorf et al., 2012; Baguette et al., 2013). For populations of aquatic species that are isolated due to habitat fragmentation, conservation resources are invested to improve connectivity through habitat restoration or barrier removal, with the intent to increase population size and improve persistence (Branco et al., 2014; Donald & Evans, 2006). It is often desirable that modifications to landscape structure will improve connectivity among habitat patches and allow for increased movement of individuals and their genes at a rate sufficient to reduce population fragmentation and vulnerability to extirpation. However, quantifying and validating the efficacy of such improvements is often challenging because it requires long-term genetic monitoring or extensive sampling of populations that may be difficult to accomplish given limited conservation resources.

While improving functional connectivity has been a common objective in restoration ecology, significant trade-offs exist for this strategy as a conservation measure. Potential positive outcomes include improving population viability through dispersal, as well as limiting genetic drift and improving gene flow and genetic diversity (i.e. genetic rescue) which can be critical for the fitness of individuals and populations of conservation concern (Baguette et al., 2013; Kovach et al., 2022; Whiteley et al., 2015). Improved gene flow can also result in the spread of adaptive alleles, which can be critical to the persistence of populations coping with changing environmental conditions due to climate or disturbance (Creech et al., 2017; Razgour et al., 2018). Barriers are sometimes used as tools to protect native species from threats to population persistence (Fausch et al., 2009; Muhlfeld et al., 2012). Thus, potential negative consequences of improving connectivity may include accelerating the spread of disease and accompanying population declines (Nobert et al., 2016), invasions by non-native species (Bell et al., 2021; With, 2002), the spread of agricultural pests (Margosian et al., 2009) and the spread of maladaptive genotypes (Fitzpatrick & Reid, 2019; Lowe & Allendorf, 2010). Additionally, connecting populations that have been separated for a significant amount of time on the evolutionary scale comes with possible trade-offs in terms of loss of genetic purity of a subpopulation, reductions in fitness due to outbreeding depression or the loss of unique evolutionary lineages and adaptive capacity (Biermann & Havlick, 2021).

Questions about the costs and benefits associated with interventions to support functional connectivity are complex and often

difficult to study experimentally because of large spatial and temporal scales and the lack of reproducibility across different landscape types (Short Bull et al., 2011). Simulation models are a valuable tool to experimentally evaluate the effects of functional connectivity on populations without the constraints of time and resources required for empirical studies (Day, Zollner, et al. 2020; Forester et al., 2023; Mims et al., 2023; Seaborn et al., 2023). Simulations that are spatially explicit and individual-based can provide insights for managing heterogeneous landscapes and the mechanistic nature of these models makes them robust to alternative scenarios of management and land use (McClane et al., 2011; Radchuk et al., 2019; Seaborn et al., 2023). Simulations also benefit from the ability to maintain a controlled environment while systematically adjusting variables and therefore provide insights about mechanisms driving landscape-level processes. When combined with empirically collected data, simulation outcomes can be used to inform data collection efforts and future management and conservation actions (e.g. adaptive management; Walters, 1986).

Landscape genetics (or riverscape genetics in riverine systems) has become an important analytical framework for understanding and measuring functional connectivity (Manel et al., 2003). The field of landscape genetics is the formal integration of spatial and genetic data, where genetic distances among individuals or populations are linked to landscape variables to make inferences or predictions about functional connectivity. Hypotheses about how the landscape affects genetic connectivity can be as simple as isolation by distance (IBD; Wright, 1943), where geographic distance is the sole predictor of genetic distance (Hutchison & Templeton, 1999). In contrast, isolation by resistance (IBR) hypotheses (McRae, 2006) involve assigning resistance values to landscape variables such as environmental resistance (e.g. gradient, temperature) or anthropogenic barriers (e.g. dams, culverts) according to their propensity to impede movement. Much attention has been given recently to the development of statistical methods for identifying the optimal IBR hypothesis for a given population (Peterman & Pope, 2021; Pless et al., 2021; Shirk et al., 2018; Vanhove & Launey, 2024). Ultimately, understanding how the landscape mediates genetic connectivity can inform strategies for restoring or improving functional connectivity among populations.

We used a landscape genetics approach combining both empirical and simulated data to explore questions of functional connectivity for a species of conservation concern, the Westslope Cutthroat Trout (*Oncorhynchus clarkii lewisi*; hereafter WCT). Our objective was to characterize the population and landscape genetics of the WCT population, and then use the results to parameterize and calibrate riverscape genetic simulations to investigate the respective roles of anthropogenic barriers and dispersal behaviour in genetic structure and diversity. This study provides an example for how long-term sampling and genetic monitoring of a population combined with landscape genetics, movement ecology and spatially explicit simulation modelling can be incorporated into the methodology, design and planning of targeted conservation efforts to understand and improve functional connectivity.

2 | METHODS

2.1 | Study system

The Sullivan Creek watershed, a tributary to the Pend Oreille River in north-eastern Washington State, United States, provides an ideal aquatic system to study how landscape resistance (landscape features that impede or promote movement) and dispersal behaviour affect functional connectivity (Figure 1). In this system, the demographic characteristics, movement and genetic structure of this resident population of WCT have been studied since 2012 in conjunction with projects focused on habitat restoration, removal of anthropogenic barriers and improving the underlying ecosystem function of the watershed (Day et al., 2018, 2021). Between 2012 and 2022, 17,871 WCT have been tagged with passive integrated transponder (PIT) tags for subsequent recapture by a combination of stream electrofishing surveys and stationary PIT antenna arrays to track growth, survival and movement of WCT across their distribution within Sullivan Creek.

2.2 | Population genetics

A total of 1582 individual WCT samples were collected at 27 locations in the Sullivan Creek watershed including the mainstem and nine tributaries during 2011, 2012 and 2013. Samples were grouped into spatial subpopulations to ensure good spatial representation and at least 30 samples per subpopulation (Figure 1). Tissues were collected from adults and subadults (fork length range 77–250 mm). Genomic DNA was extracted from tissue samples and genotyped by the Molecular Genetics Lab at the Washington Department of Fish and Wildlife. Genotypes consisted of 19 microsatellites (MSATs; see Small et al., 2012 for loci ID and genotyping protocols) with 247 alleles (mean = 13 per locus, range = 5–21) and 187 single nucleotide polymorphisms (SNPs). The SNPs were generated in previous efforts employing RADseq methodology, screened with a GT-seq protocol and corrected to account for SNPs that exhibited patterns consistent with selection (Campbell et al., 2015; Small et al., 2015, 2017). Population allele frequencies were calculated after removing hybrids (fish with at

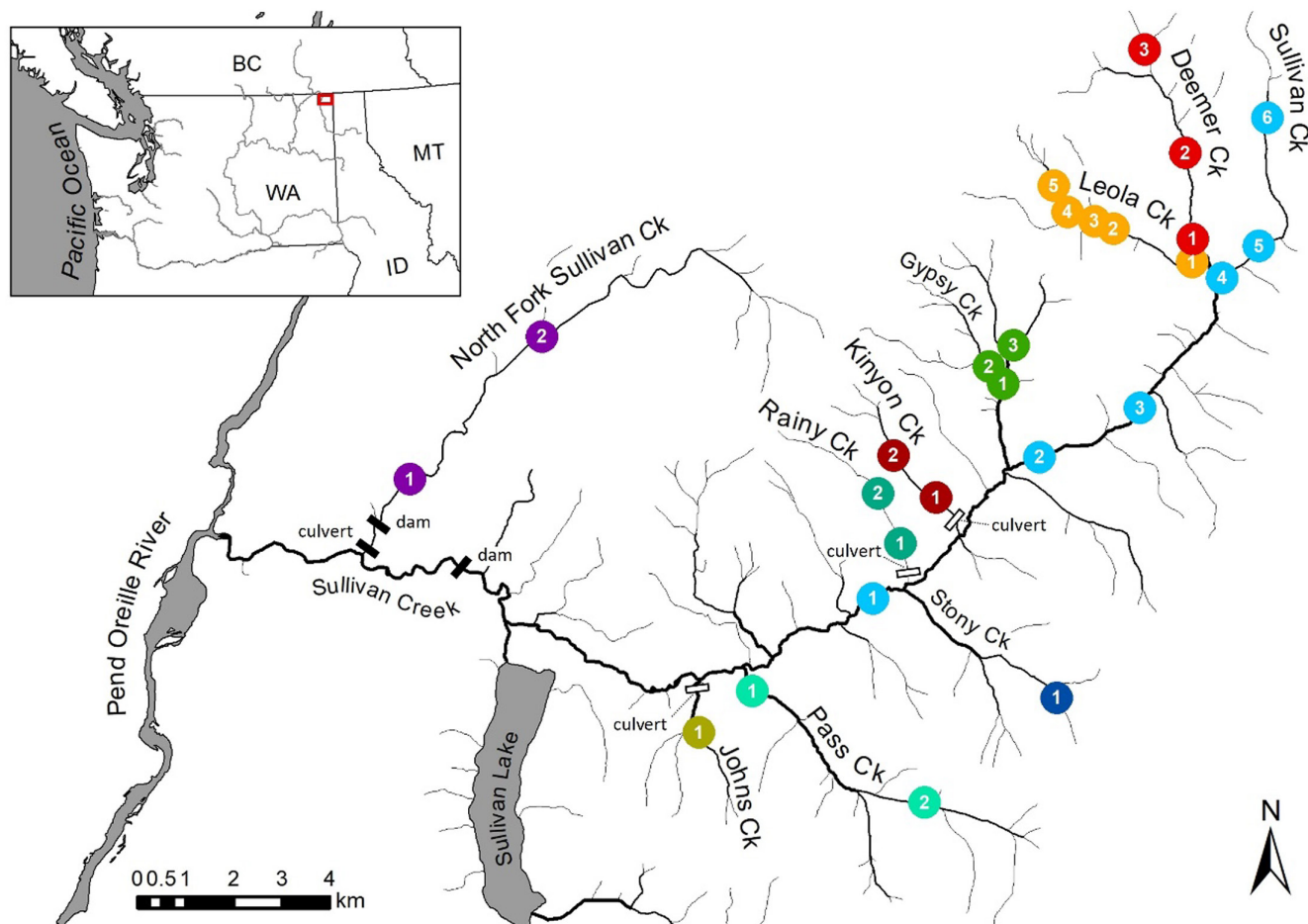


FIGURE 1 Study Area. Locations of 27 subpopulations of Westslope Cutthroat Trout (circles) and anthropogenic barriers in the Sullivan Creek Basin in north-eastern Washington State, United States. Sampled streams ($n=10$) are indicated by uniquely coloured circles with names for each stream indicated on the map. Numbers within each circle represent the spatial delineation of subpopulations within each stream. Unsourced streams contained insufficient numbers of Westslope Cutthroat Trout for genetic analysis. Location of complete and partial barriers to movement (as dam or culverts) shown with bars (black—complete or passability = 0; hollow—partial with passability between 0 and 1).

least one Rainbow Trout (*Oncorhynchus mykiss*) allele at diagnostic loci; Small et al., 2017), and missing loci were infilled using the population mode.

Population genetics analyses were conducted on both the empirical and simulated data sets for both MSAT and SNP marker sets at the scale of the sampled locations displayed in Figure 1. Simulated data sets (see description of simulation below) were subsampled to match the sampling plan used for the collection of the empirical data (i.e. same spatial patches and number of captures). To evaluate genetic diversity, we calculated per locus and average expected heterozygosity, H_s (Nei, 1987), and compared it to overall gene diversity, H_t (Nei, 1987), using the R package 'hierfstat' (Goudet & Jombart, 2022). For genetic structure, we used the same package to calculate global genetic differentiation using both global F_{ST} (Nei, 1987), and Jost's genetic differentiation estimator, D_{est} (Jost, 2008). These two metrics measure the extent to which genetic variation between populations exceeds that within populations (i.e. genetic structure), and we included both metrics to improve confidence in results. We conducted principal component analysis (PCA) using the R package 'ade4' (Jombart, 2008) and visualized results using the R package 'ade4' (Dray & Dufour, 2007). We calculated population-specific F_{ST} (Weir & Goudet, 2017; Weir & Hill, 2002) for the sampled patches using the 'betas' function in 'hierfstat'. Population-specific F_{ST} describes the genetic structure of a subpopulation relative to the overall population by comparing the likelihood of two individuals in a subpopulation matching alleles to the likelihood of an individual matching alleles with a random individual from the overall population. We selected seven patches across three streams that were representative of the watershed by including three reaches from the mainstem, two from a tributary above an anthropogenic barrier and three from a tributary with no anthropogenic barrier: Sullivan 1, 3 and 6; North Fork 1 and 2; and Leola 1, 3 and 4. (Figure 1). We plotted results for both the MSAT and SNP data sets and the IBD and IBR landscape hypotheses.

2.3 | Riverscape genetics

We conducted a landscape (or here, riverscape) genetic analysis for WCT in Sullivan Creek to both understand how landscape features affect the genetic structure and to inform the movement modules in the simulation model. First, we calculated pairwise genetic distances among the 27 sampled subpopulations within mainstem Sullivan Creek and its tributaries (Figure 1). Samples were grouped a priori into subpopulations based on tributary and proximity, and all subpopulations contained at least 28 samples (total $n = 1138$); enough to ascertain accurate population allele frequencies (Hale et al., 2012). Because measures of genetic distance can vary with population attributes (Jost, 2008; Shirk et al., 2017) and to provide additional confidence in results, we included results from three differentiation metrics calculated with R package 'diveRsity' (Keenan et al., 2013): Nei's G_{ST} (Nei, 1973), Jost's D_{est} (Jost, 2008) and Hedrick's G_{ST} (Hedrick, 2005). We compared results from analyses for MSAT and

SNP data individually and selected the genetic distance metric that resulted in the highest R^2 value during landscape genetics model selection.

We used the CircuitScape method (i.e. circuit theory; McRae et al., 2013) in R package 'resistanceGA' (Peterman, 2018) to test for relationships between landscape features and genetic distance. ResistanceGA combines a machine learning genetic algorithm with maximum likelihood population effects regression (MLPE; Clarke et al., 2002) with the effective distances (here, calculated with CircuitScape) and pairwise genetic distance to optimize the resistance surface. We identified four variables as likely contributors to resistance to movement for WCT: riverine distance, and effective distances considering gradient, elevation and anthropogenic barriers (Davis et al., 2018). Spatial raster layers representing elevation and gradient (i.e. environmental resistance) were extracted from a digital elevation model retrieved from the US Geological Survey National Map (usgs.gov/the-national-map-data-delivery). Resistance values due to anthropogenic barriers (dams and culverts) were estimated based on the authors' expert opinions, and represented as 1.0 for an impassable barrier, and as 0.0 in all other raster cells. We tested two versions of the anthropogenic barrier variable, one that included only complete barriers to upstream movement and one that included both complete and partial barriers (Figure 1). Because ResistanceGA optimizes resistance surfaces from a suite of potential transformations, no transformations to input surfaces were made prior to running the optimization algorithm.

We used a hierarchical approach to model selection using ResistanceGA. First, we optimized each surface individually to test for any significant relationship with genetic distance. We then included only those significant variables in a multi-surface optimization (Peterman, 2018). To reduce model complexity, we removed any variables that contributed less than 5% to the composite surface and repeated the analysis. Final model selection was performed based on Akaike's information criteria (AICc, Akaike, 1998; Row et al., 2017).

2.4 | Model overview

To simulate the demogenetics of the WCT population, we used the modeling framework Cost Distance MetaPOPulation version 1.44 (CDMetaPOP; Landguth et al., 2017). CDMetaPOP is an individual-based, spatially explicit, forward-in-time landscape genetics simulation framework. In CDMetaPOP, landscapes are composed of patches established across continuous space, which are in turn populated with individuals. Individuals within a patch share a common environment (e.g. spatial location, temperature, carrying capacity), but experience life-history processes individually. These processes include growth, mortality, maturation, reproduction, dispersal and migration, among others. Gene flow patterns emerge from the dispersal and reproduction of individual fish, as alleles assigned to offspring follow Mendelian inheritance. Below we provide a brief description of the parameterization of the WCT application of CDMetaPOP. For a full list of parameters, see Table S1.

2.4.1 | Landscape parameters

We divided the portion of the Sullivan Creek basin occupied by WCT into 415 patches averaging 200 m in length. All field data were spatially assigned to this same network of patches. Each patch was assigned a carrying capacity (K) reflective of the estimated abundance of WCT, calculated as follows: we first estimated abundance for 130 patches that were sampled in the field from 2014 to 2017 using multi-pass depletion surveys (Zippin, 1956). Abundance estimates were based on a closed population capture–recapture removal model fit using maximum likelihood. We then developed a linear regression model using the R package ‘lme4’ (Bates et al., 2015) to predict WCT abundance in those patches that were not sampled, using temperature, subwatershed (Upper/Middle/Lower Sullivan Creek) and Brook Trout abundance as explanatory variables.

Water temperature for each patch was calculated separately for the spawning period (1st June to 1st October; 123 days) and the overwintering period (2nd October to 31st May; 242 days). To determine patch-specific temperatures, we used data from a network of 34 HOBO Tidbit and ProV2 temperature sensors deployed throughout the study area from 1 July 2013 to 23 July 2015. We then used PCA to model the spatial and temporal variation in stream temperature data. For a full description of this analysis, see Day et al. (2018).

2.4.2 | Individual parameters

Individual WCT were parameterized according to empirical data collected throughout the Sullivan basin. Growth was based on the von Bertalanffy curve (Von Bertalanffy, 1938) adjusted for temperature (Landguth et al., 2017). We first estimated von Bertalanffy growth function parameters based on an empirical sample of 303 WCT otoliths collected from 2015 to 2017. From these parameters, we used the temperature-adjusted von Bertalanffy curve to simulate a size–age data set based on patch temperature. We then conducted Bayesian analysis using JAGS (Plummer, 2003) and the R package ‘R2jags’ (Su & Yajima, 2015) to fit the parameters used by CDMetaPOP to calculate annual incremental growth as a function of fish age and patch temperature.

Probability of maturation followed a logistic function of size (Downs et al., 1997):

$$p = \exp(A + B * \text{Length}) / (1 + \exp(A + B * \text{Length})) \quad (1)$$

where p equals the probability of reproductive maturity, and the intercept (A) and slope (B) were derived from empirical data. This function was calculated separately for females ($A=0.06$, $B=-8.09$) and males ($A=0.13$, $B=-20.28$). Fecundity was modelled as an exponential function of size (Bowerman, 2013):

$$\text{Number of eggs} = C * \exp(\text{Length} * D) \quad (2)$$

where parameters C (9.56) and D (0.018) were derived from empirical data.

2.4.3 | Movement parameters

We used the resistance surface generated from the landscape genetics analysis and observations of annual movements to fit a *resistance* dispersal distribution by calculating the cumulative resistance associated with each observed movement. We likewise fit a *dispersal distance* distribution by calculating the riverine distance associated with each observed movement. This process was repeated for observed riverine distances moved. The distribution of annual dispersal movement was based on 847 mark–recapture observations collected from WCT during electrofishing surveys from 2012 to 2019. For each observation, riverine distance was calculated (IBD) and cumulative resistance between the patch endpoints was derived from the composite resistance surface produced by ResistanceGA (IBR). We used the R package ‘fitdistrplus’ (Delignette-Muller & Dutang, 2015) to test and compare the fit of several potential dispersal distributions to these sets of distance and resistance movements (Figure S1). For riverine distance, the Pareto distribution provided the best fit based on the Kolmogorov–Smirnov goodness-of-fit statistic ($KS=0.083$). For resistance, both the Pareto and Weibull distributions provided reasonable fits, but the Weibull provided a better fit for the tail portion of the movement distribution (shape=0.52, scale=7306, $KS=0.172$). To implement movement during simulations, we created a matrix of pairwise riverine distance values and a matrix of cumulative resistance values using the software UNICOR (Landguth et al., 2012), and an asymmetrical barrier distance matrix was calculated using the directional graph option in UNICOR (Mims et al., 2019). Depending on the landscape hypothesis, dispersing individuals would then select and move to a dispersal patch following a random draw from the corresponding dispersal distribution.

The primary source of mortality for WCT in the simulations was a size-structured density dependence model (Day, Landguth, et al., 2020; Stige et al., 2019). This model of density dependence implements both intra- and inter-cohort competition within each patch. Individuals within a given size class compete to survive within the carrying capacity allocated to that size class given an ideal size distribution for the population. Inter-cohort competition followed the adult-dominance model, whereby survival of smaller size classes was regulated by the density of individuals of larger size classes (Sandlund et al., 1991).

2.4.4 | Study design

We developed four primary simulation scenarios in a fully crossed study design to investigate the effects of two factors: anthropogenic barriers to movement (Figure 1) and dispersal distribution (Figure S1). For the first factor, anthropogenic barriers to movement (i.e. dams and culverts) were either absent (Models 1 and 3 defined as ‘no barriers’) or present (Models 2 and 4 defined as ‘barriers’) in each scenario. These anthropogenic barriers include four culverts and a diversion dam on tributaries to Sullivan Creek and a dam on the lower reach of Sullivan Creek (Figure 1). For the second factor, two dispersal distributions

were selected as they related to the landscape resistance surface. In the simulations, dispersal events were drawn either from an empirical dispersal distribution derived from fitting annual movement observations to a leptokurtic distribution described above (Models 1 and 2 defined as 'restricted') or a more mobile (i.e. linear) dispersal distribution that resulted in greater movement than was empirically observed in the population (Models 3 and 4 defined as 'mobile'). Thus, Model 1 evaluated empirical dispersal in a landscape without barriers, Model 2 assessed empirical dispersal in a landscape with barriers, Model 3 included mobile dispersal in a landscape without barriers and Model 4 implemented mobile dispersal in a landscape with barriers (Figure 2).

Within each of the four scenarios, we ran simulations representing both an isolation-by-distance (IBD) and the empirically derived isolation-by-resistance (IBR) hypothesis to explore how the added landscape effects genetic outcomes. Dispersal in these respective simulations was parameterized according to either the IBD (based on riverine distance) or IBR (based on landscape genetics analysis) hypothesis by fitting observed distance and resistance movements, respectively, to an empirical distribution. In the case of the empirically derived movement scenarios, observed 'distance' and 'resistance' movements were fit to a Pareto or Weibull distribution, respectively, and the respective distribution was used to draw individual movement probabilities between patches from a cost matrix of pairwise distances or resistances. For the mobile dispersal scenarios, individual movement was based on a linear relationship between either distance or resistance and probability of movement between patches. Finally, to investigate effects of marker type, all simulations included both SNP and MSAT markers with

starting spatial allele frequencies derived from the empirical population. We ran 10 replicates of the four simulation scenario times the two landscape hypotheses (8 total combinations) for 500 years (including a demographic burn-in period of 25 years), comparing and evaluating the outcome variables at year 500.

3 | RESULTS

3.1 | Empirical genetics

Expected genetic heterozygosity (H_e) was 0.55 and 0.26 for MSATs and SNPs, respectively, while overall gene diversity (H_d) was 0.72 and 0.37 respectively (Table 1). Overall F_{ST} was lower for MSATs (0.23) than for SNPs (0.31) while the opposite pattern was true for D_{est} (MSATs=0.38, SNPs=0.16). PCA on the empirical data indicated spatial population structure for both the MSAT and SNP data sets (Figure 3). For both markers, the North Fork tributary, which is separated from mainstem Sullivan Creek by two anthropogenic barriers to upstream movement and a natural steep gradient reach, displayed tight groupings indicating population structure. Similarly, the headwater reaches of Sullivan Creek, which contain no anthropogenic barriers but do contain steep gradient reaches, displayed population structure relative to the individuals sampled from the mainstem. Other separations from the mainstem population included Rainy Creek and Kinyon Creek, which similarly contain no complete anthropogenic barriers but contain steep gradient reaches and partial barriers.

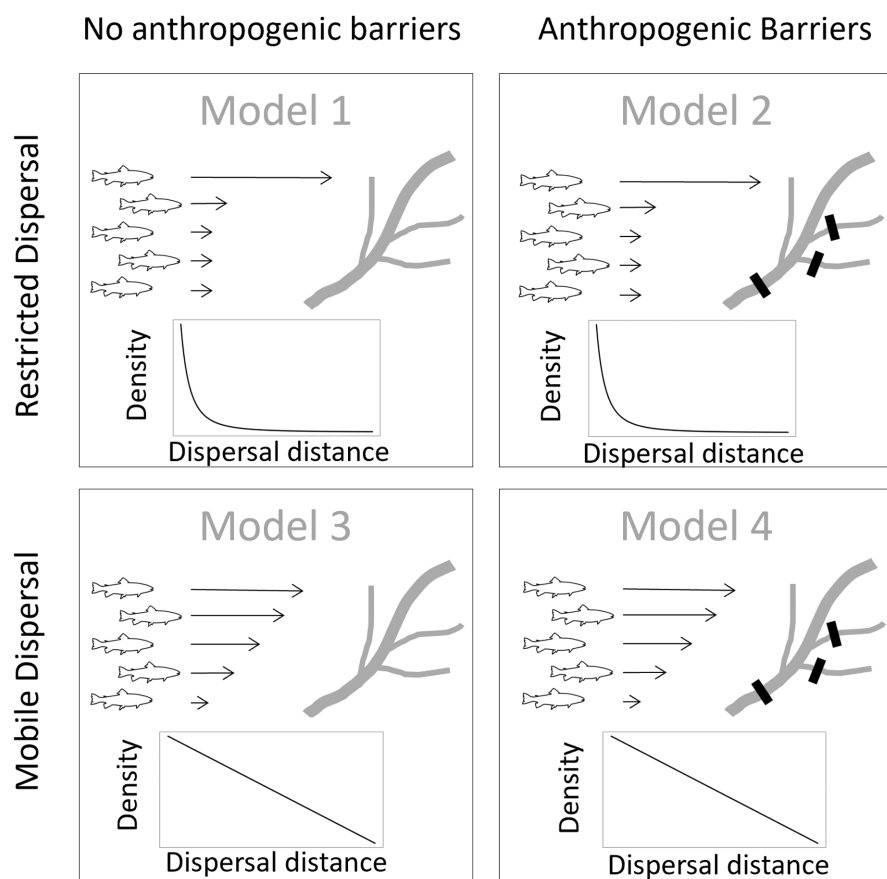


FIGURE 2 Factorial study design for landscape genetics simulations of Westslope Cutthroat Trout in Sullivan Creek, Washington, United States. Black bars represent the presence of partial and complete anthropogenic barriers to upstream movement. Observed fish movements were fit to leptokurtic dispersal distributions for the restricted movement scenario and a linear relationship between dispersal and raw resistance/distance units was used for the mobile scenario. Graphs show the probability density function for the respective dispersal distribution, and arrows represent relative distances moved by fish as determined by the relevant dispersal distribution presented in the graphs.

TABLE 1 Global genetic summary statistics.

	MSATs				SNPs			
	H_o	H_t	F_{ST}'	D_{est}	H_o	H_t	F_{ST}'	D_{est}
Empirical	0.55	0.72	0.23	0.38	0.26	0.37	0.31	0.16
Model 1	0.55 (0.006)	0.72 (0.005)	0.25 (0.009)	0.40 (0.012)	0.27 (0.003)	0.37 (0.002)	0.28 (0.005)	0.14 (0.002)
Model 2	0.55 (0.012)	0.72 (0.006)	0.25 (0.009)	0.40 (0.010)	0.27 (0.004)	0.37 (0.004)	0.29 (0.006)	0.15 (0.003)
Model 3	0.72 (0.008)	0.73 (0.007)	0.01 (0.001)	0.03 (0.003)	0.39 (0.002)	0.39 (0.003)	0.01 (0.001)	0.01 (0.001)
Model 4	0.70 (0.006)	0.72 (0.007)	0.04 (0.003)	0.10 (0.006)	0.35 (0.003)	0.37 (0.003)	0.07 (0.003)	0.04 (0.002)

Note: Genetic statistics for the empirical genetic data set and 10 replicates at year 500 for each model with means ($\pm$ SD) for observed heterozygosity (H_o), overall gene diversity for total population (H_t), fixation index (F_{ST}') and genetic differentiation (D_{est}) for 19 microsatellite (MSAT) markers and 187 single nucleotide polymorphism (SNP) markers. Subsampling of the simulated populations was used to match the sample size and locations of the empirical samples ($N=1004$). Model 1 is empirical movement without barriers, Model 2 is empirical movement with barriers, Model 3 is mobile movement without barriers and Model 4 is mobile movement with barriers.

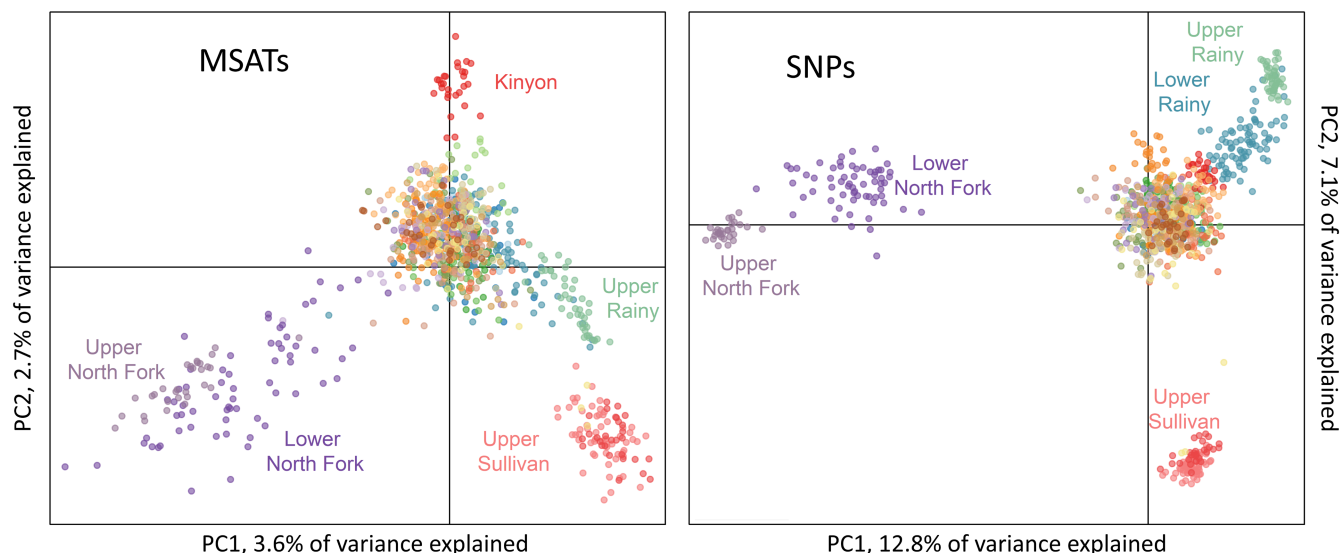


FIGURE 3 Principal component analysis. Results from a principal component analysis on empirical genotypes of individual Westslope Cutthroat Trout plotted on the first two principal components using microsatellite (MSAT; left panel) and single nucleotide polymorphism (SNPs; right panel) marker sets. Colours correspond to spatial locations of 27 sample locations throughout the Sullivan Creek study area in north-eastern Washington State, United States (as seen in Figure 1).

3.2 | Riverscape genetics

Of the four variables that we evaluated as contributors to resistance to movement, elevation and gradient were significant predictors of genetic distance between WCT subpopulations and therefore included in the best model of landscape resistance. Neither riverine distance nor anthropogenic barriers improved model fit and therefore were not included in the final resistance model. Because anthropogenic barriers were not included, we henceforth refer to this model of landscape resistance as 'environmental resistance'. Overall, elevation made the greatest contribution to the environmental resistance surface followed by gradient (Table S2). Across genetic distance metrics, MSATs showed greater contributions from gradient as compared to SNPs. Also, the contribution of each variable was dependent on the genetic distance metric, with elevation making the greatest contribution when using Nei's G_{ST} , followed by

Jost's D and Hedrick's G_{ST} (Table S2). Jost's D , which we used to parameterize movement distributions for the final model, resulted in a marginal R^2 of .42 and a conditional R^2 of .89. When we conducted single-surface optimizations, the optimal transformation for the elevation and gradient layers were the inverse Ricker and reverse Ricker transformations, respectively (Figure S2). Comparisons of the performance of the genetic distance metrics across marker types are shown in Table S2.

3.3 | Simulation results

3.3.1 | Global genetic structure

Mean values across 10 replicates of 500-year simulations with restricted movement (Models 1 and 2) reflected the global genetic

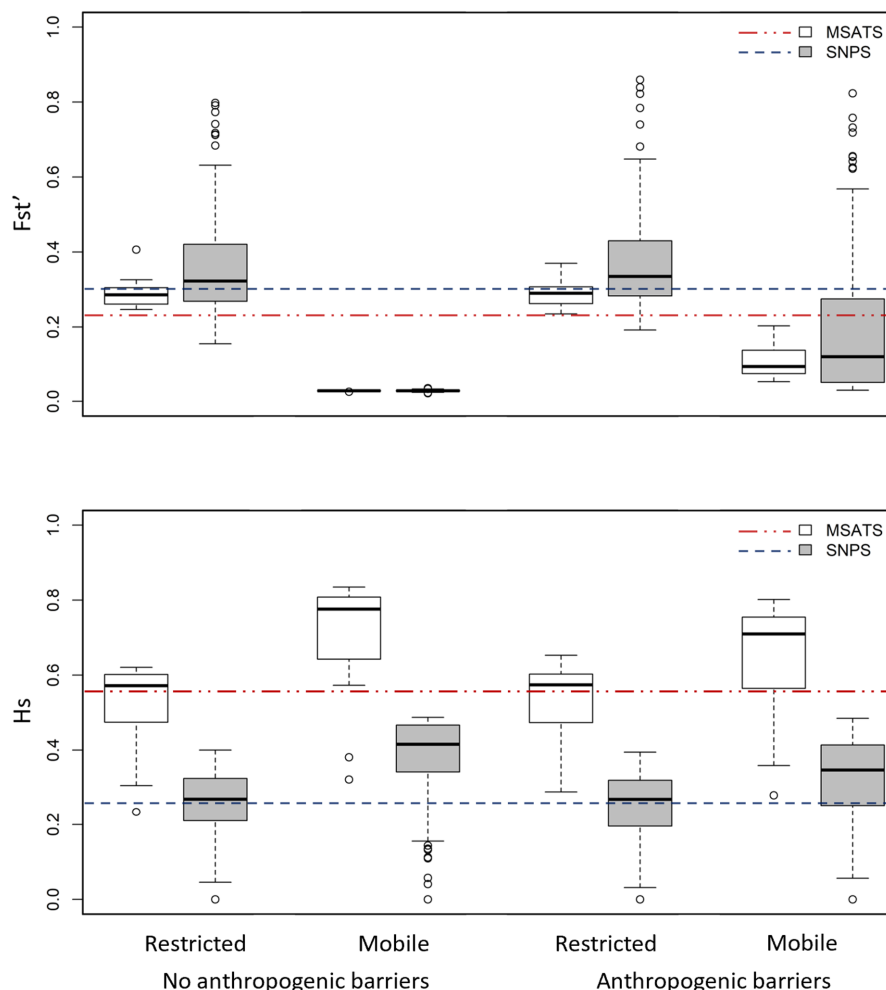


FIGURE 4 Global genetic structure. Per-locus genetic structure (F_{ST}') and expected heterozygosity (H_s) of four simulation scenarios of Westslope Cutthroat Trout in north-eastern Washington State, United States, that vary by movement distribution (restricted vs. mobile) and inclusion of anthropogenic barriers. Dashed lines represent mean values for the empirical data set. Simulations were subsampled prior to analysis to match the sampling design from the empirical data. The restricted movement distribution was fit to observed fish movements.

diversity and structure of the contemporary data set (Table 1, Figure 4). Removing anthropogenic barriers (Model 1) had only minor effects on global population structure for the restricted movement scenarios (Figure 5). Both simulation scenarios with mobile movement (Models 3 and 4) produced higher genetic diversity and lower genetic structure than the empirical movement counterparts. Adding barriers to the mobile scenarios increased genetic structure, as populations were panmictic only within barrier groupings (Figure 4). Results from the two marker sets (MSATs and SNPs) were qualitatively consistent across all four models in terms of genetic diversity and structure. As in the empirical data set, F_{ST}' values were lower and H_s values higher when using MSAT data compared to SNP data (though D_{est} values were higher, Figure S3).

3.3.2 | Spatial genetic structure

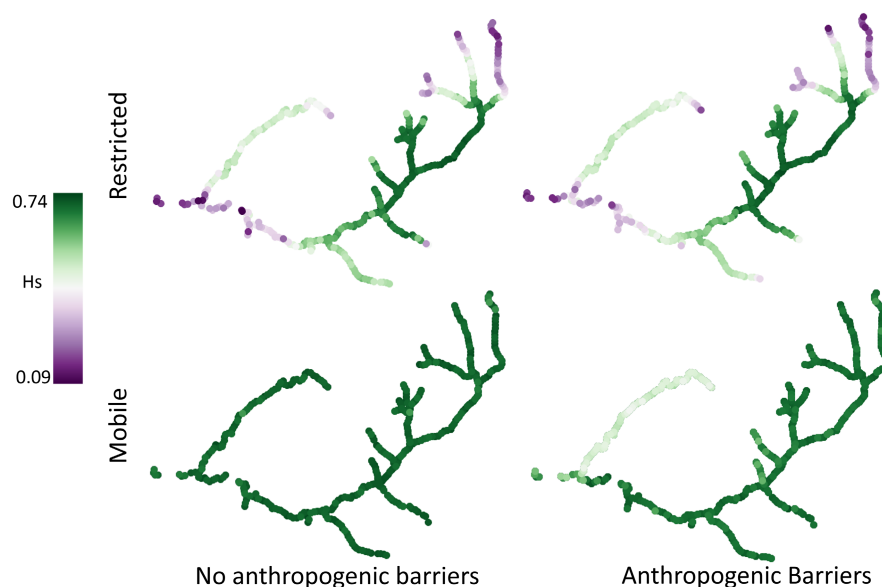
PCA on the simulated data (Figure 6) shows varying spatial genetic structure across model scenarios. Similar to the empirical genetic data, the simulations parameterized with empirical movement data show greater genetic structure for the North Fork tributary and Sullivan Creek headwaters relative to the rest of the watershed. Mobile simulation scenarios display panmixia within barrier groups

and population structure between barrier groups. Spatial plots of heterozygosity support these results and demonstrate how for empirical movement scenarios, patches in distal locations displayed lower heterozygosity than those located centrally (Figure 5).

3.3.3 | Population-specific genetic differentiation

Population-specific F_{ST} generally increased with distance from the mainstem of Sullivan Creek for both marker sets and both IBD and IBR simulations (Figure 7). The MSAT and SNP analyses resulted in the same qualitative results across patches, though SNPs averaged higher F_{ST} values than MSATs. Simulations with fish dispersal based on IBR versus IBD also displayed similar results, with some key differences. First, IBR resulted in higher values of F_{ST} than IBD where population genetic structure was identified on the landscape, particularly in upper tributaries but with the exception of the Upper North Fork patches where F_{ST} values were similar. Second, IBR produced higher values than expected for the lower North Fork patch when compared to the upper North Fork patch, especially in the MSAT analysis. Finally, the highest F_{ST} values in the IBD simulations were in the North Fork tributary, whereas the highest F_{ST} values in the IBR simulations were in the Upper Sullivan and Upper Leola

FIGURE 5 Spatial genetic diversity. Distribution of H_s across four simulation scenarios of Westslope Cutthroat Trout in north-eastern Washington, United States. Factors that were varied included use of either an empirically derived, restricted dispersal distribution versus a mobile dispersal distribution and the inclusion or exclusion of dams and culverts that impede upstream movements.



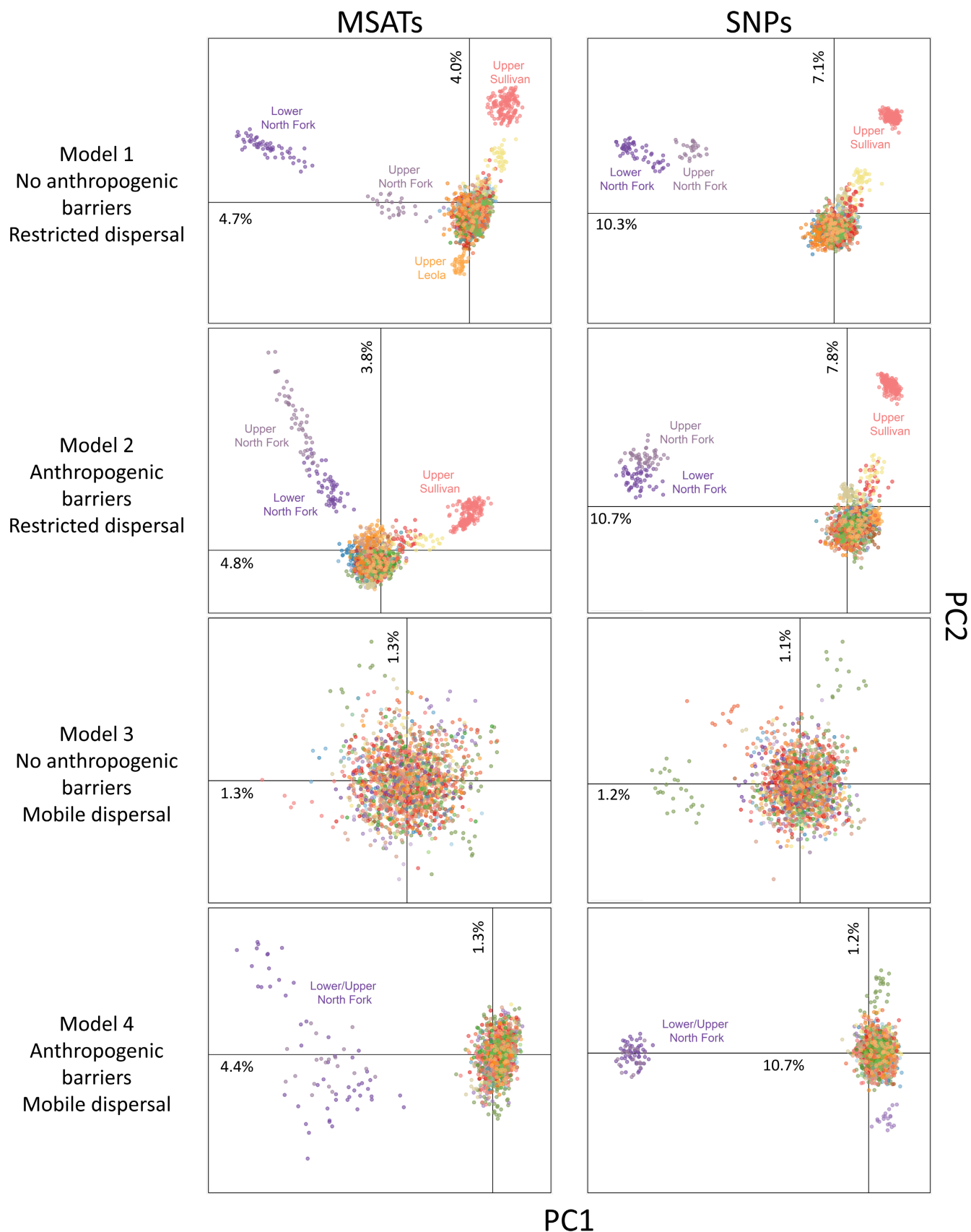
tributaries. These values from the IBR simulations corresponded with isolated collections of WCT in Sullivan and Leola creeks above high environmental resistance reaches due to effects of gradient and elevation.

4 | DISCUSSION

We evaluated the relative contributions of different components of functional connectivity (movement behaviour, environmental resistance and anthropogenic barriers) in a resident population of Westslope Cutthroat Trout using a long-term behavioural, demographic and genetic data set with spatially explicit landscape genetics simulations that model species' interactions across the landscape. With the empirical data revealing patterns of restricted dispersal and genetic differentiation in distal tributary populations, our landscape genetics analyses for this population of WCT indicated that environmental resistance to movement (i.e. gradient, elevation) was a stronger driver of population structure than existing anthropogenic barriers. While the relative recency of anthropogenic barriers on the landscape may dampen the expected patterns of population structure over a longer timeframe, outcomes from individual-based simulations supported this result. The simulations produced similar patterns of genetic structure with and without anthropogenic barriers in the empirical dispersal scenarios over a timespan that exceeds the presence of anthropogenic barriers in the study system (i.e. 500 years). Below, we discuss insights from this work that highlight the role of functional landscape connectivity while considering landscape structure, dispersal behaviour and genetic connectivity when considering management and restoration actions to conserve populations. Such an approach could increase the success of management actions focused on the conservation and reconnection of populations by incorporating knowledge of both the landscape structure and animal behaviour that together regulate functional connectivity.

4.1 | Role of dispersal in landscape genetics

Landscape genetics has been underutilized for riverscapes (Davis et al., 2018) yet was critical to understanding relative drivers of population structure and functional connectivity in this study, as well as developing simulations to explore mechanisms driving genetic outcomes for WCT. This approach was strengthened by first using empirical data to optimize a model of landscape resistance and then fitting a dispersal distribution based on observed resistance (rather than distance) movements to inform movement distributions in the simulations. Empirical landscape genetics is an established field of study, but the fitting of movement distributions to empirical resistance data for simulations, to our knowledge, has not previously been done. Because the isolation-by-resistance (IBR) model outperformed the isolation-by-river-distance (IBD) model (McRae, 2006), our simulations produced more accurate projections of genetic structure than if the IBD model alone were used to simulate dispersal. Specifically, the landscape resistance model identified high resistance in high-elevation tributaries and low resistance in the watershed mainstem. While this may be an intuitive result, fitting a dispersal distribution to resistance instead of distance resulted in higher structure among those regions than in the IBD simulations where the dispersal distribution was fit to distances alone. Critically, these improvements to projections in the IBR simulations highlighted the important role of fish movement in functional connectivity more than in the IBD simulations. These results also underlie the importance of the interaction between dispersal and gene flow for evaluating the conservation status of animal populations and resulting interactions between landscape structure and population genetic structure (Baguette et al., 2013). Because dispersal is not necessarily indicative of gene flow, our approach of deriving a resistance surface and dispersal distribution from genetic data is likely to be more useful for simulating landscape effects on gene flow than analyses based purely on observations of animal movements.



4.2 | Role of simulation modelling in riverscapes

Simulations can play a critical role in determining the long-term evolutionary impacts of anthropogenic barriers to movement and their

removal on aquatic species (Blair et al., 2012; Seaborn et al., 2023). This is particularly relevant where anthropogenic barriers have not been in place long enough to detect effects on genetic structure, especially when the behaviours (e.g. short-range dispersal) and life

FIGURE 6 Principal component analysis. Results from a principal component analysis on genetic scores of simulated Westslope Cutthroat Trout for 27 subpopulations (represented by colour) throughout Sullivan Creek in north-eastern Washington State, United States. Microsatellite (MSAT; left column) and single nucleotide polymorphism (SNP; right column) genotypes are displayed for four simulation models. The percent of variation explained by each PC is labelled on each plot axis. Model 1=restricted (i.e. empirically derived) movement distribution with no anthropogenic barriers, Model 2=restricted movement distribution with anthropogenic barriers, Model 3=mobile movement distribution with no anthropogenic barriers and Model 4=mobile movement distribution with anthropogenic barriers.

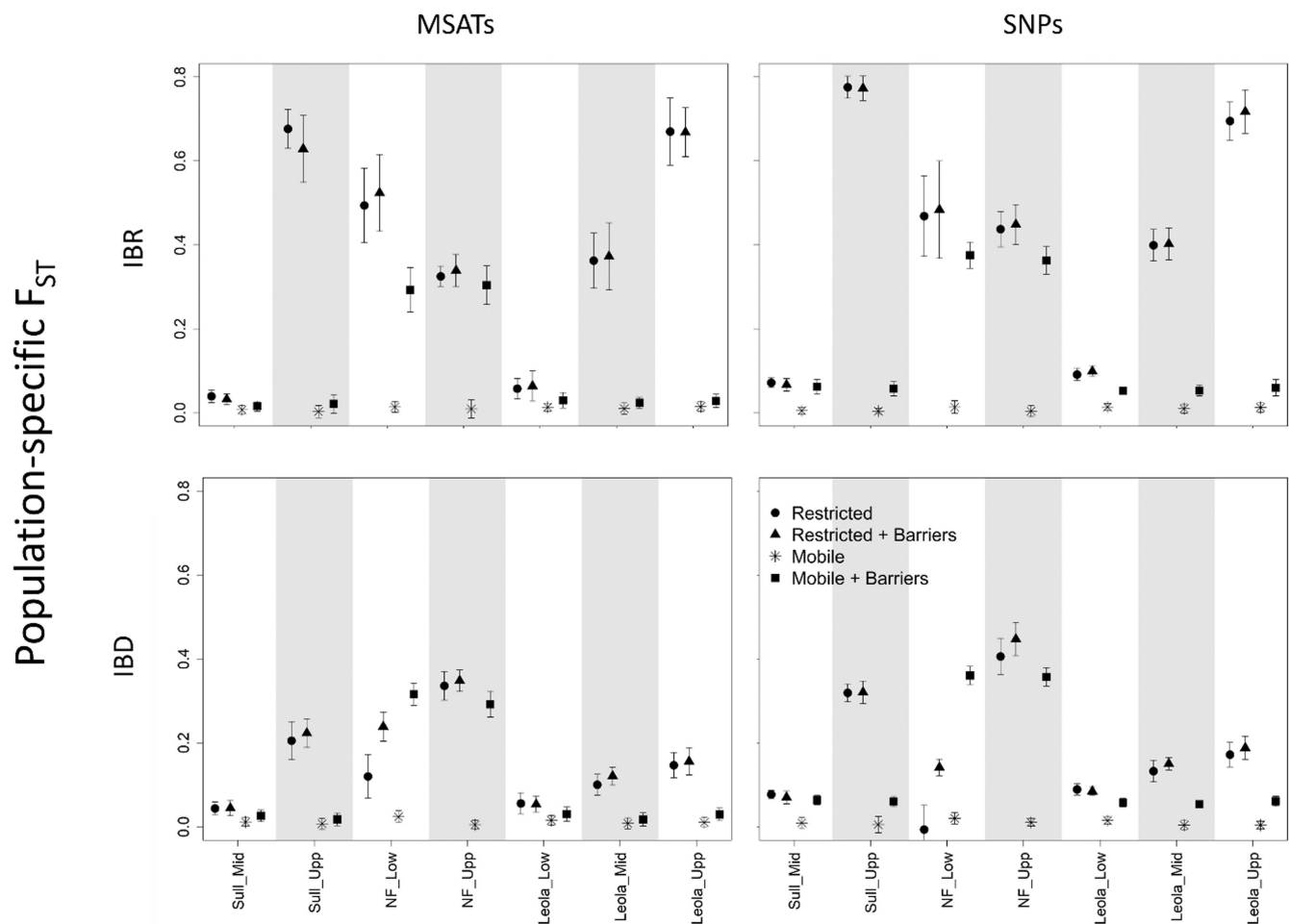


FIGURE 7 Population-specific genetic differentiation. Mean ($\pm$ SD) population-specific F_{ST} across 10 replicates from a simulated population of Westslope Cutthroat Trout in Sullivan Creek, Washington State, United States. Results from seven sampling locations are presented in order from downstream to upstream (left to right) for each of three streams (NF=North Fork, Sull=mainstem Sullivan Creek, Leola=Leola Creek), and were selected to be representative of the watershed as a whole, including mainstem reaches, tributary reaches and reaches above and below anthropogenic barriers. Only the North Fork tributary is isolated from mainstem Sullivan Creek by anthropogenic barriers to upstream movement. IBD represents an Isolation-by-distance hypothesis, and IBR represents an Isolation-by-resistance hypothesis identified through model selection.

histories (e.g. resident) expressed can require a longer response time to make genetic barriers detectable (Landguth et al., 2010). Thus, simulations can be a critical tool for understanding long-term impacts and informing best management practices. In our case, empirical genetic data were combined with a long-term capture-recapture data set that yielded robust estimates of movement and demographic parameters, allowing for the simulation experiments to focus on landscape and management factors.

When using simulation modelling for making predictions or future forecasting, care must be taken to avoid spuriousness in model

results due to overfitting of model parameters when calibrating the model to match empirical patterns (Anderson et al., 2001; Cushman & Landguth, 2010). In contrast, simulations may be most useful when used to compare relative outcomes among competing hypotheses (Gallagher et al., 2021; Shirk et al., 2012). This approach tests the ability of simulated mechanisms to reproduce observed patterns and puts more emphasis on effect sizes than statistical tests (White et al., 2014). When used as both a problem-solving tool and a forecasting tool (Starfield, 1997), simulations can play a crucial role in understanding how patterns of animal behaviour and landscape

configuration drive eco-evolutionary processes at the population scale. In our study, we used empirical patterns of fish dispersal and landscape resistance to produce process-driven simulation results. This mechanistic approach allows for greater confidence in results from the simulation experiments, in which we used a factorial design to compare relative outcomes of competing hypotheses in our WCT system.

4.3 | Comparing marker types

While we found differences between our marker sets, the goal of this work was not to compare performance of MSATs and SNPs. We acknowledge that much has already been done to compare the performances of marker types, though inclusion of both markers here serves to contribute to the body of knowledge on the topic and to improve confidence in our conclusions. For example, Lemopoulos et al. (2019) compared the use of 4876 SNPs and 16 MSATs for brown trout (*Salmo trutta*) and found that the two sets performed similarly for estimating genetic structure. There are many reasons why we could have observed differences here, but most notably, our empirical and simulated data included relatively few SNPs (187) and genetic structure in our system is not as obvious as in Lemopoulos et al. We also acknowledge the potential for bias due to sampling and sampling size. The genetic structure identified using both metrics in the simulations generally matched expectations due to barriers to movement and environmental resistance derived from empirical observations, though SNPs showed stronger patterns of structure than MSATs. For heterozygosity, it is notable that despite having different absolute values, the two marker sets displayed similar qualitative patterns across the different simulation scenarios.

4.4 | IBD versus IBR

Clear differences emerged between simulations with dispersal movement parameterized using distance-only observations (Isolation-by-distance; IBD) and resistance-only observations (Isolation-by-resistance; IBR). We remind readers that the IBR landscapes were derived from the optimized landscape genetics results and only included elevation and gradient as predictors. IBD simulations assumed that riverine distance was the only factor affecting where an individual is likely to disperse, and this resulted in lower genetic structure between mainstem and tributary populations than would be expected from the empirical data. Resident fish populations such as the one simulated here often show fine-scale genetic structure among isolated subpopulations in tributaries (Carlsson et al., 1999; Kanno et al., 2011; Pilger et al., 2017), and incorporating environmental resistance due to gradient and elevation in the IBR simulations resulted in a more realistic genetic landscape. Accuracy in this regard can be critical for any project prioritizing populations for conservation actions such as barrier removal or translocations (Muhlfeld et al., 2012).

It is possible that the IBR simulations resulted in overestimation of genetic structure between tributaries behind barriers and the mainstem. While CDMetaPOP implemented effects of anthropogenic barriers in an asymmetric design (i.e. one-way barriers), the algorithm we used to optimize landscape resistance is symmetrical (Peterman, 2018). Environmental landscape resistance is therefore bidirectional even though there likely is less resistance for a fish to move downstream than upstream. This may be the cause for the counterintuitive simulation result where the lower North Fork tributary shows greater genetic distance with the overall population than it does with the upper North Fork tributary. Because streams are both linear in nature and naturally sloped, asymmetrical barriers can have a stronger effect than in other ecosystems, making this area of research important for future riverscape genetics research and application.

4.5 | Environmental landscape resistance and anthropogenic barriers

In this study, we considered the effects of environmental resistance to movement (e.g. natural gradient barriers, elevation) and anthropogenic barriers. We found environmental resistance to movement to be a stronger driver of genetic structure than the existing anthropogenic barriers. Most likely due to restricted movement across the riverscape in this study system, the removal of anthropogenic barriers did not significantly alter the future genetic structure of the population, particularly in tributaries to the mainstem where environmental resistance was highest. This result suggests that when high environmental resistance and anthropogenic barriers co-occur, their effects on connectivity can be difficult to tease apart. Anthropogenic barriers are a relatively recent phenomenon, and natural resistance may describe genetic structure better than anthropogenic barriers (Deiner et al., 2007; Fraik et al., 2021) because they have resulted in the evolution of unique genetic lineages while maintaining genetic connectivity in the downstream direction. These interactions of historical barrier placement on the outcome of genetic structure observed should be further explored.

In contrast to the hypothesis that environmental resistance drives genetic structure in this study system, it is possible that the observed pattern of restricted dispersal developed as a result of the placement of those barriers followed by selection for resident life history. That is, an outstanding question that emerged from this study is what shaped the resident life history of the study system? Restricted movement has long been a commonly held concept for stream-dwelling fish species (Gerking, 1959), though others have challenged this characterization for salmonids (Gowan et al., 1994). Individual WCT have been shown to move great distances regardless of watershed size, and while such movements may be uncommon, these long-distance movements may have a significant impact on genetic exchange and overall population structure (Gresswell & Hendricks, 2007; Mills & Allendorf, 1996). If the restricted dispersal distribution was recently shaped by anthropogenic barriers, it is

possible that a more mobile life-history strategy would emerge following barrier removal.

Our results contribute to the ongoing discussion about the costs and benefits of stream barriers and barrier removal. Because streams are linear in nature, anthropogenic barriers can play a significant role in limiting movement and gene flow in aquatic species. Barriers such as culverts and dams fragment populations and block the passage of migratory fish returning to natal grounds for spawning. Construction of these barriers has resulted in declines of migratory fish species (Lawrence et al., 2016), loss of genetic diversity (Zarri et al., 2022), changes to community structure (Freedman et al., 2014), altered geomorphology and flow regimes (Ligon et al., 1995) and loss of Indigenous and commercial fishing practices (Montgomery, 2003). Subsequently, removal of dams and culverts has garnered much attention and support in recent decades (Bednarek, 2001; Doyle et al., 2003, but see Jones et al. (2021) for a review of barrier usage for the prevention of the spread of invasive species). Studies have quantified the benefits of barrier removal for improving gene flow and connectivity among populations (Landguth et al., 2010; Prunier et al., 2023), and many methods for identifying and prioritizing barrier removals have been developed (e.g. Doyle et al., 2003; Walter et al., 2021). Our results demonstrate that in addition to weighing potential costs and benefits in terms of connectivity, gene flow, invasive species and disease spread, decisions surrounding barrier removal should be made with consideration for both existing and historical movement distributions and life-history strategies within the context of each study system. Such questions present fruitful exploration avenues for simulation modelling in aquatic systems and their use in support of management decisions.

4.6 | Limitations and uncertainties

All simulation studies carry uncertainty in model parameters and must make simplifying assumptions to focus on key processes driving the model system. In our study, results were largely driven by the patterns of movement based on empirically derived distributions that were potentially biased based on factors like sampling locations and size of fish sampled. Furthermore, salmonid movement patterns can vary by age (Young, 2011) and fish may be more mobile as juveniles, which are more difficult to capture. As with any capture-recapture study, it can also be difficult to detect long-distance movements, especially when individuals disperse outside of the study area. Finally, the existing genetic structure of this population of Westslope Cutthroat Trout may have been affected by a history of introductions and impacts from non-native species—effects that are not easily captured in landscape genetics analysis. While our study presents a step forward in using landscape resistance to parameterize simulated dispersal movements, these limitations present avenues for future research on the role of movement and life-history strategies in functional connectivity of aquatic systems (Neville et al., 2006; Ostergren & Nilsson, 2012).

4.7 | Conservation implications

Our main finding of the work was that for this resident Westslope Cutthroat Trout population, the effects of existing anthropogenic barriers on population structure were redundant with effects of restricted dispersal associated with the underlying environmental resistance. While the simulations did not show evidence of barrier removal being effective at connecting isolated populations in this particular system, we do not suggest that such an outcome will always be the case. Rather, we emphasize here that consideration of movement behaviour with landscape resistance can be critical to inform decisions around restoration of connectivity. Our approach provides an example of how extensive field sampling combined with landscape genetics and spatially explicit simulation modelling can be incorporated into the methodology, design and planning of targeted conservation efforts to understand and improve functional connectivity. Tools used to optimize barrier removal in streams often aim to maximize structural connectivity and habitat availability but do not have the ability to consider dispersal behaviour or genetic outcomes for the target species (e.g. Branco et al., 2014; O'Hanley et al., 2013). By simulating the genetic effects of barrier removal, our study describes a useful approach for practitioners to prioritize places where animal behaviour is most likely to contribute to improved gene flow and functional connectivity.

AUTHOR CONTRIBUTIONS

AB, RKS, CCD and ELL were involved in conceptualization. MS and RKS were involved in data curation. CCD, RKS, ELL and AB were involved in methodology. CCD, DA and ELL were involved in formal analysis. CCD, DA, SRC, IFG, KEZ, RKS, MS, ELL and MS were involved in investigation. CCD, DA, IFG, RKS and SRC were involved in visualization. AB, RKS, CCD and ELL were involved in funding acquisition. RKS, KEZ and SRS were involved in project administration. CCD, ELL, RKS and MJF were involved in supervision. CCD and ELL were involved in software. CCD, DA and SRC were involved in Writing—original draft. CCD, ELL, RKS, SRC, MJF, IFG, KEZ and SRS were involved in writing—review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Genetic distance matrices and environmental surfaces used in the landscape genetics analyses, simulation input files and genotypes are available at: DOI:<https://doi.org/10.5061/dryad.vmcvndn25> (Day et al. 2024).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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